Homework Assignment 1: Slab-Geometry Boltzmann Equation

Tal Ramot

August 10, 2026

1 Introduction & Case's Method

This report presents the solutions to Homework Assignment 1, which focuses on the monoenergetic, steady-state Boltzmann transport equation in slab geometry for an infinite, homogeneous, isotropic medium with an isotropic plane delta-source at $x = 0$. The transport equation is written as:

$$\mu \frac{\partial \psi(x, \mu)}{\partial x} + \psi(x, \mu) = \frac{c}{2} \int_{-1}^{1} \psi(x, \mu') d\mu' + \frac{1}{4\pi} \delta(x) \tag{1}$$

where $\psi(x, \mu)$ is the angular flux, $c \in [0, 1)$ is the scattering ratio, and μ is the cosine of the angle of the particle's velocity with the x-axis.

In 1953, Kenneth Case analyzed this equation by searching for separation-of-variables solutions of the form:

$$\psi_\nu(x, \mu) = e^{-x/\nu} \phi_\nu(\mu) \tag{2}$$

Substituting this ansatz into the homogeneous transport equation leads to the eigenvalue problem:

$$(\nu - \mu)\phi_\nu(\mu) = \frac{c\nu}{2} \int_{-1}^{1} \phi_\nu(\mu') d\mu' \tag{3}$$

Normalizing the eigenfunctions such that $\int_{-1}^{1} \phi_\nu(\mu') d\mu' = 1$, we obtain:

$$(\nu - \mu)\phi_\nu(\mu) = \frac{c\nu}{2} \tag{4}$$

Depending on the value of the eigenvalue ν, two cases arise:

1. **Discrete Spectrum ($\nu \notin [-1, 1]$):** Since $\nu - \mu \neq 0$ for all $\mu \in [-1, 1]$, we can divide directly by $\nu - \mu$, yielding the discrete eigenfunctions:

$$\phi_0^\pm(\mu) = \frac{c\nu_0}{2(\nu_0 \mp \mu)} \tag{5}$$

   Integrating these over $[-1, 1]$ and applying the normalization condition yields the transcendental characteristic equation for the discrete eigenvalues $\pm\nu_0$:

$$1 = c\nu_0 \text{arctanh}(1/\nu_0) \tag{6}$$

   For $c \in (0, 1)$, there is exactly one pair of real, positive roots $\pm\nu_0$ with $\nu_0 > 1$.

2. **Continuous Spectrum ($\nu \in [-1, 1]$):** In this case, the term $\nu - \mu$ vanishes at $\mu = \nu$. Thus, we cannot divide by $\nu - \mu$ in the sense of ordinary functions. Instead, in the theory of distributions (generalized functions), the general solution to the equation is:

$$\phi_\nu(\mu) = \frac{c\nu}{2}\mathcal{P}\frac{1}{\nu - \mu} + \lambda(\nu)\delta(\mu - \nu) \tag{7}$$

where $\mathcal{P}$ represents the Cauchy principal value distribution, $\delta(\mu-\nu)$ is the Dirac delta function, and $\lambda(\nu)$ is the dispersion function. Integrating $\phi_\nu(\mu)$ over $\mu \in [-1, 1]$ and setting the integral to 1 determines $\lambda(\nu)$:

$$\lambda(\nu) = 1 - c\nu\text{arctanh}(\nu) \tag{8}$$

1.1 Intuition Behind the Continuous Spectrum and Generalized Functions

The presence of the Dirac delta function $\delta(\mu-\nu)$ and the principal value term $\mathcal{P}\frac{1}{\nu-\mu}$ in the continuous eigenfunctions $\phi_\nu(\mu)$ can feel counter-intuitive.

Physically, the continuous spectrum represents the singular behavior of particles that travel without undergoing collisions (uncollided transport) or undergo highly directional scattering. A particle traveling at a specific direction $\mu = \nu$ represents a singular distribution in angle, which is modeled by the Dirac delta $\delta(\mu - \nu)$. The principal value distribution $\mathcal{P}\frac{1}{\nu-\mu}$ accounts for the strong coupling between nearby angles due to scattering, where the singularity is balanced symmetrically.

However, when we look at the scalar flux $\phi(x)$, which is the angle-integrated quantity:

$$\phi(x) = 2\pi \int_{-1}^{1} \psi(x, \mu) d\mu \tag{9}$$

these angular singularities are integrated out. Under integration, the distribution $\phi_\nu(\mu)$ acts as a linear functional mapping to a constant:

$$\int_{-1}^{1} \phi_\nu(\mu) d\mu = 1 \tag{10}$$

This leads to the transient component of the scalar flux:

$$\phi_{\text{tr}}(x) = \frac{1}{2}\int_{0}^{1} \frac{e^{-|x|/\nu}}{N_\nu} d\nu \tag{11}$$

where:

$$N_\nu = \nu \left[\lambda^2(\nu) + \frac{\pi^2 c^2 \nu^2}{4}\right] \tag{12}$$

Crucially, **no distributions or generalized functions remain in the scalar flux integral**. The integrand is a regular, smooth function of ν on the open interval $(0, 1)$. For any $x > 0$, as $\nu \to 0$, the exponential term $e^{-|x|/\nu}$ decays extremely rapidly, making the integrand vanish at 0. As $\nu \to 1$, the dispersion function $\lambda(\nu) \to -\infty$, which causes the normalization factor $N_\nu \to \infty$ and the integrand to vanish at 1. Thus, the numerical integration is highly well-behaved.

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2 Question 1: Exact Scalar Flux Analysis

The exact scalar flux is computed as:

$$\phi(x) = \phi_{\text{as}}(x) + \phi_{\text{tr}}(x) \tag{13}$$

where $\phi_{\text{as}}(x) = \frac{e^{-|x|/\nu_0}}{2N_0^+}$ is the asymptotic component, and $\phi_{\text{tr}}(x)$ is the transient component. The discrete normalization factor is:

$$N_0^+ = \frac{c}{2}\nu_0^3 \left[\frac{c}{\nu_0^2 - 1} - \frac{1}{\nu_0^2}\right] \tag{14}$$

For ν_0, we can choose to compute it using:

- **Numerical Root-Finding:** Solving $\frac{\text{arctanh}(y)}{y} = \frac{1}{c}$ for $y = 1/\nu_0 \in (0, 1)$ using a bisection method, which is extremely stable.
- **Analytic Approximation:**

$$\nu_0 \approx \frac{1}{\sqrt{1 - c^{p(c)}}} \tag{15}$$

where:

$$p(c) = 2.47412 + \frac{0.00363081}{c^2} - 0.0352458c + \frac{0.557498}{c} \tag{16}$$

2.1 Question 1a: Asymptotic and Transient Components

[Insert plots of asymptotic and transient components for each c, and the relative contributions $\phi_{\text{tr}}(x)/\phi(x)$ and $\phi_{\text{as}}(x)/\phi(x)$ as a function of x. Add physical discussion here.]

2.2 Question 1b: Classical Diffusion Solution

The classical diffusion equation (in units of mean free path, where $\Sigma_t = 1$) for a homogeneous, isotropic medium with an isotropic plane delta-source is given by:

$$-D\frac{d^2\phi_{\text{class}}(x)}{dx^2} + (1 - c)\phi_{\text{class}}(x) = \delta(x) \tag{17}$$

Using the standard P1 diffusion coefficient $D = 1/3$, the Green's function solution is:

$$\phi_{\text{class}}(x) = \frac{\sqrt{3}}{2\sqrt{1 - c}} e^{-\sqrt{3(1-c)}|x|} \tag{18}$$

[Insert comparison plots with the classical diffusion solution. Discuss the differences, including the behavior near the source ($x \to 0$) and the asymptotic behavior ($x \to \infty$).]

2.3 Question 1c: Asymptotic Diffusion Solution

The asymptotic diffusion approximation matches the diffusion length to the exact discrete transport eigenvalue ν_0, i.e., $L = \nu_0$, which corresponds to an asymptotic diffusion coefficient $D_{\text{as}} = (1 - c)\nu_0^2$. The governing equation is:

$$-D_{\text{as}}\frac{d^2\phi_{\text{as,diff}}(x)}{dx^2} + (1 - c)\phi_{\text{as,diff}}(x) = \delta(x) \tag{19}$$

which yields the Green's function solution:

$$\phi_{\text{as,diff}}(x) = \frac{1}{2(1-c)\nu_0} e^{-|x|/\nu_0} \tag{20}$$

[Insert comparison plots with the asymptotic diffusion solution. Discuss how the asymptotic diffusion solution compares to both the classical diffusion solution and the exact transport asymptotic component $\phi_{\text{as}}(x)$.]

2.4 Question 1d: Relative Error of Diffusion Approximations

The relative error of each diffusion approximation relative to the exact transport solution is defined as:

$$\text{Relative Error (\%)} = \frac{|\phi_{\text{approx}}(x) - \phi_{\text{exact}}(x)|}{\phi_{\text{exact}}(x)} \times 100\% \tag{21}$$

[Insert relative error plots as a function of x. Add physical discussion comparing the accuracy of the classical and asymptotic diffusion approximations across the different values of c. Analyze why the error behaves as it does near the source and far from it.]

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3 Question 2: Numerical Code for Diffusion Equation

To solve the time-independent diffusion equation with a delta-source at $x = 0$ numerically, we exploit the symmetry of the medium and integrate over the half-domain $x \in [0, a]$. Since the delta-source is located at $x = 0$, the domain $x > 0$ is source-free, governed by the homogeneous equation:

$$-D\frac{d^2\phi(x)}{dx^2} + (1-c)\phi(x) = 0 \tag{22}$$

3.1 Analytical Conversion to a First-Order System

We convert this second-order ODE to a linear system of two first-order ODEs. Let:

$$y_1(x) = \phi(x), \quad y_2(x) = \frac{d\phi(x)}{dx} \tag{23}$$

This yields the first-order system:

$$\frac{d}{dx}\begin{pmatrix} y_1 \\ y_2 \end{pmatrix} = \begin{pmatrix} y_2 \\ \frac{1-c}{D} y_1 \end{pmatrix} \tag{24}$$

3.2 Modelling the Delta Source

The delta source is never discretised. Integrating the governing equation $-D\phi''(x)+(1-c)\phi(x) = \delta(x)$ from $-\epsilon$ to $+\epsilon$ and taking the limit $\epsilon \to 0$ yields the jump condition

$$-D\left[\phi'(0^+) - \phi'(0^-)\right] = 1 \tag{25}$$

By symmetry of the infinite-medium solution ($\phi(x) = \phi(-x) \implies \phi'(0^-) = -\phi'(0^+)$), this reduces to a boundary condition at the origin:

$$\phi'(0^+) = -\frac{1}{2D}, \qquad \text{equivalently} \qquad J(0^+) = -D\phi'(0^+) = \frac{1}{2} \tag{26}$$

that is, half the source neutrons stream in each direction. This is the answer to the question of how a delta-function source should be modelled: symmetry converts it into a prescribed current, the problem is solved on the half-line $x \geq 0$, and the singularity never enters the numerical scheme.

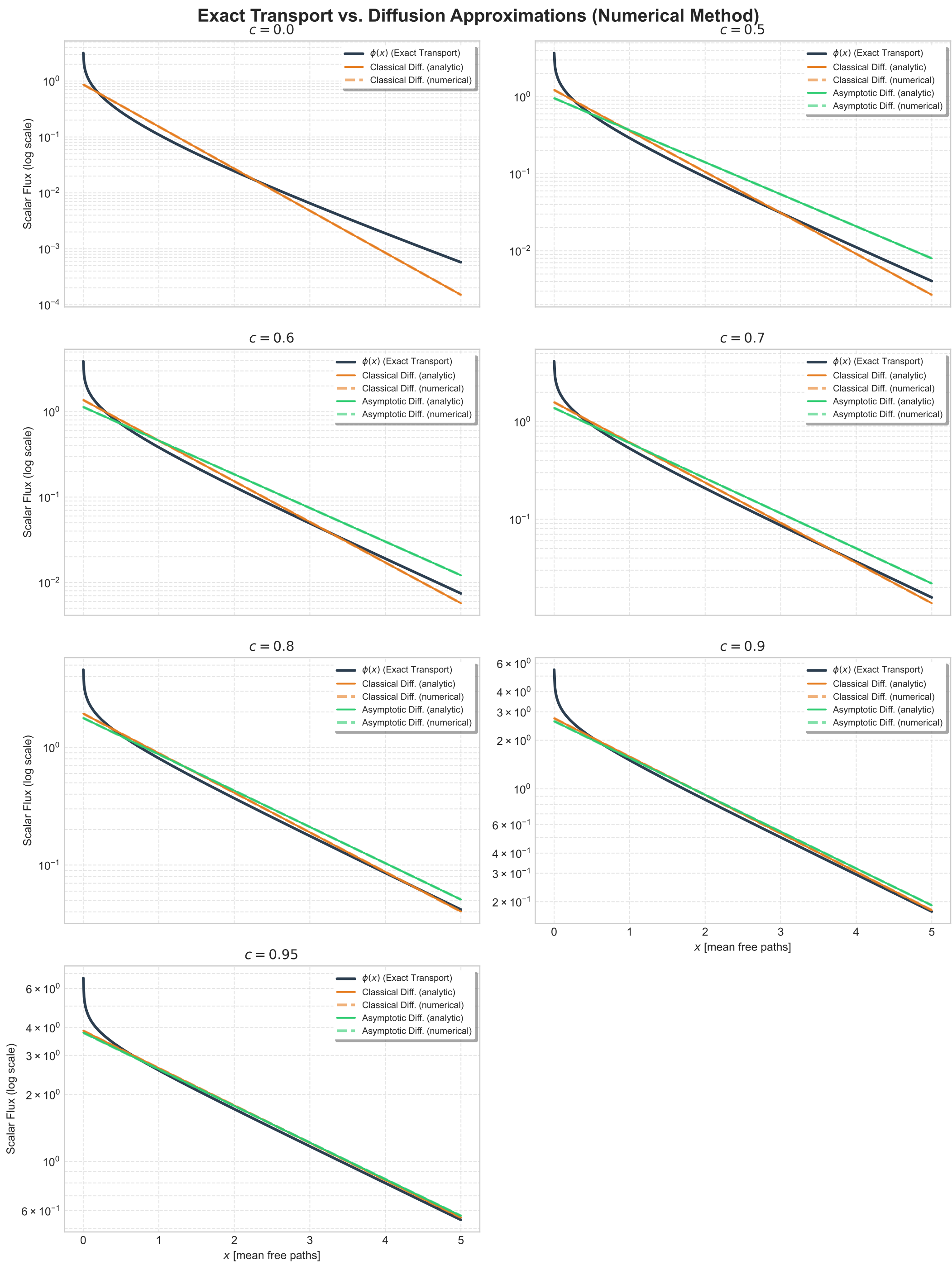

Figure 1: Exact scalar flux compared with the classical and asymptotic diffusion approximations, for $c = 0, 0.5, 0.6, 0.7, 0.8, 0.9, 0.95$. Each approximation is shown both as its analytic closed form and as the numerical solution of Question 2; the two coincide to within 2×10^{-8} % and are indistinguishable at this scale.

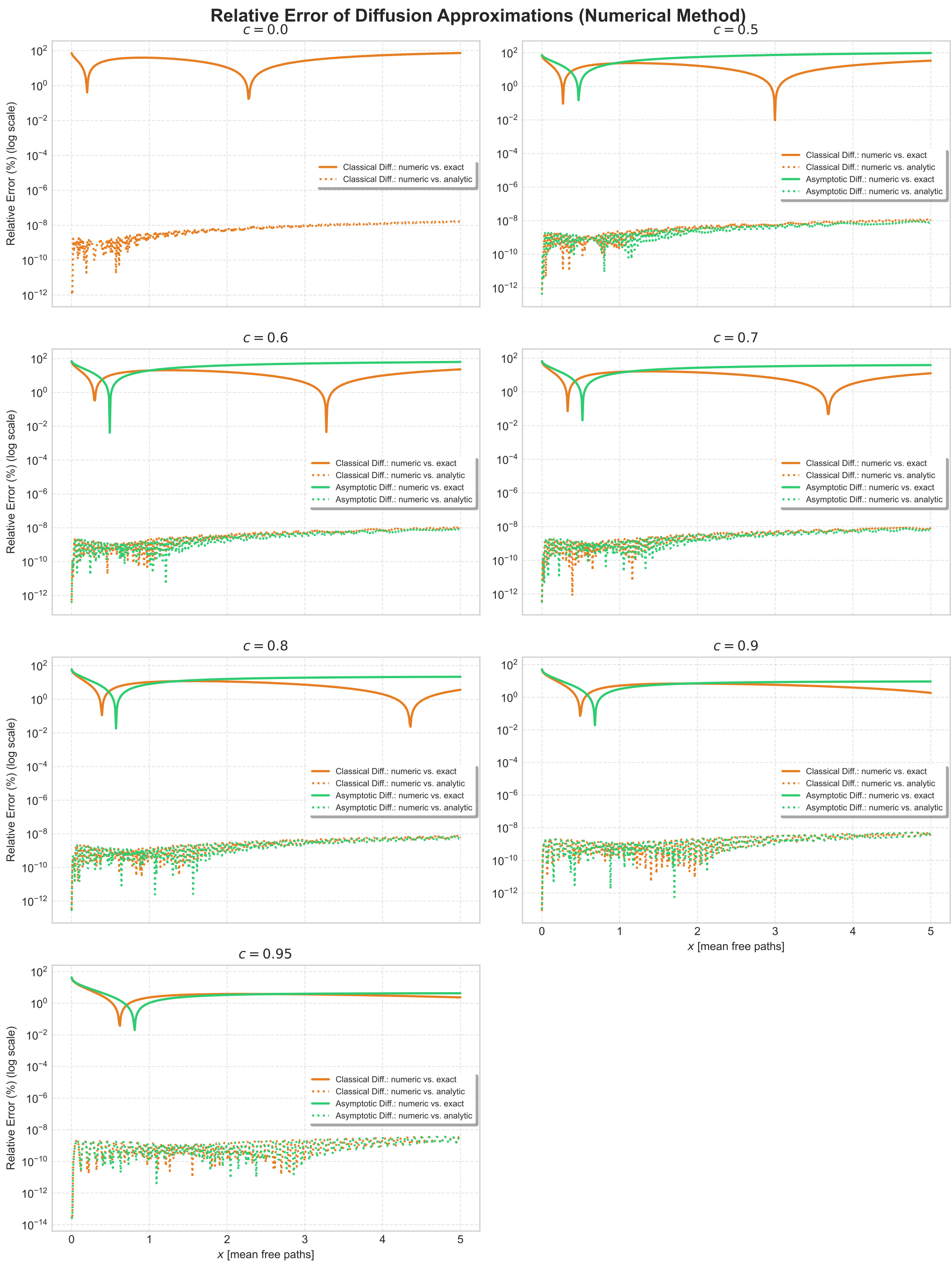

Figure 2: Two relative errors per approximation. The solid curves compare the numerical solution with the *exact transport* solution, and so measure the error of the diffusion approximation itself (Question 1d). The dotted curves compare it with the *analytic diffusion* solution, and so measure the error of the solver (Question 2). The two differ by roughly ten orders of magnitude: the discretisation error is negligible against the modelling error, so the solid curves may be read as properties of the approximation rather than of the code.

3.3 The Outer Boundary

The outer boundary is a truncation of an infinite medium, not a physical surface. Imposing a zero-flux condition $\phi(a) = 0$ would force the flux to vanish where the true Green's function is small but non-zero, producing a 100 % relative error at $x = a$ that does not shrink as the solver tolerance is reduced. We therefore impose the radiation (Robin) condition

$$\phi'(a) = -\kappa\phi(a), \qquad \kappa = \sqrt{\frac{\Sigma_a}{D}} \tag{27}$$

which the decaying solution $e^{-\kappa x}$ satisfies identically. This annihilates the growing mode $e^{+\kappa x}$, so the truncation is exact for any a; we use $a = 10/\kappa$.

3.4 Solution by a Single Integration

Away from the source the equation is linear and homogeneous, so its solution is exactly proportional to its starting amplitude: $y(x; s) = s\, y(x; 1)$. No root-finding is required. We integrate once from a towards 0 — the direction in which the solution grows, which is the numerically stable one — starting from the radiation condition with unit amplitude, $[y_1(a), y_2(a)]^T = [1, -\kappa]^T$, and then rescale the result by

$$\text{scale} = \frac{-1/(2D)}{y_2(0)} \tag{28}$$

so that the source condition $-D\phi'(0) = 1/2$ is satisfied exactly.

3.5 Both Diffusion Approximations

Classical and asymptotic diffusion are the same equation with different coefficients. Matching the closed form $\phi = e^{-\kappa|x|}/(2D\kappa)$ to Case's asymptotic mode $\phi = e^{-|x|/\nu_0}/(2(1-c)\nu_0)$ requires $\kappa = 1/\nu_0$ and hence

$$\Sigma_a = 1 - c, \qquad D_{\text{classical}} = \frac{1}{3}, \qquad D_{\text{asymptotic}} = (1 - c)\,\nu_0^2 \tag{29}$$

so a single solver covers both by choice of D.

3.6 Comparison with the Analytical Solution

The numerical solution reproduces the analytic diffusion Green's function to a maximum relative error of 2×10^{-8} % for both approximations at $c = 0.5, 0.7, 0.9$, and conserves neutrons to $\Sigma_a \int \phi\, dx = 0.99998807$; the residual is the physical tail of the Green's function beyond the truncation, not solver error. Both comparisons appear in Figures 1 and 2 above, alongside the exact transport solution.

Because a shooting method has no mesh spacing to refine — the output grid does not control accuracy — convergence is demonstrated by tightening the integrator tolerance. Figure 5 shows the relative L_2 error falling from 9×10^{-6} at `rtol` $= 10^{-4}$ to 1.3×10^{-12} at `rtol` $= 10^{-11}$, tracking the requested tolerance with unit slope.

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4 Question 3: Critical Half-Thickness and Critical Radius

This section computes $a/2$ and R_c for $1.02 < c < 2$ from the exact-transport (asymptotic) criticality relations, part 3(a), and from classical diffusion with the Marshak and Mark boundary conditions, parts 3(b) and 3(c). Parts 3(d) and 3(e), the asymptotic diffusion approximation with the exact and the modified-Marshak $\ell_0(c)$, are not covered here.

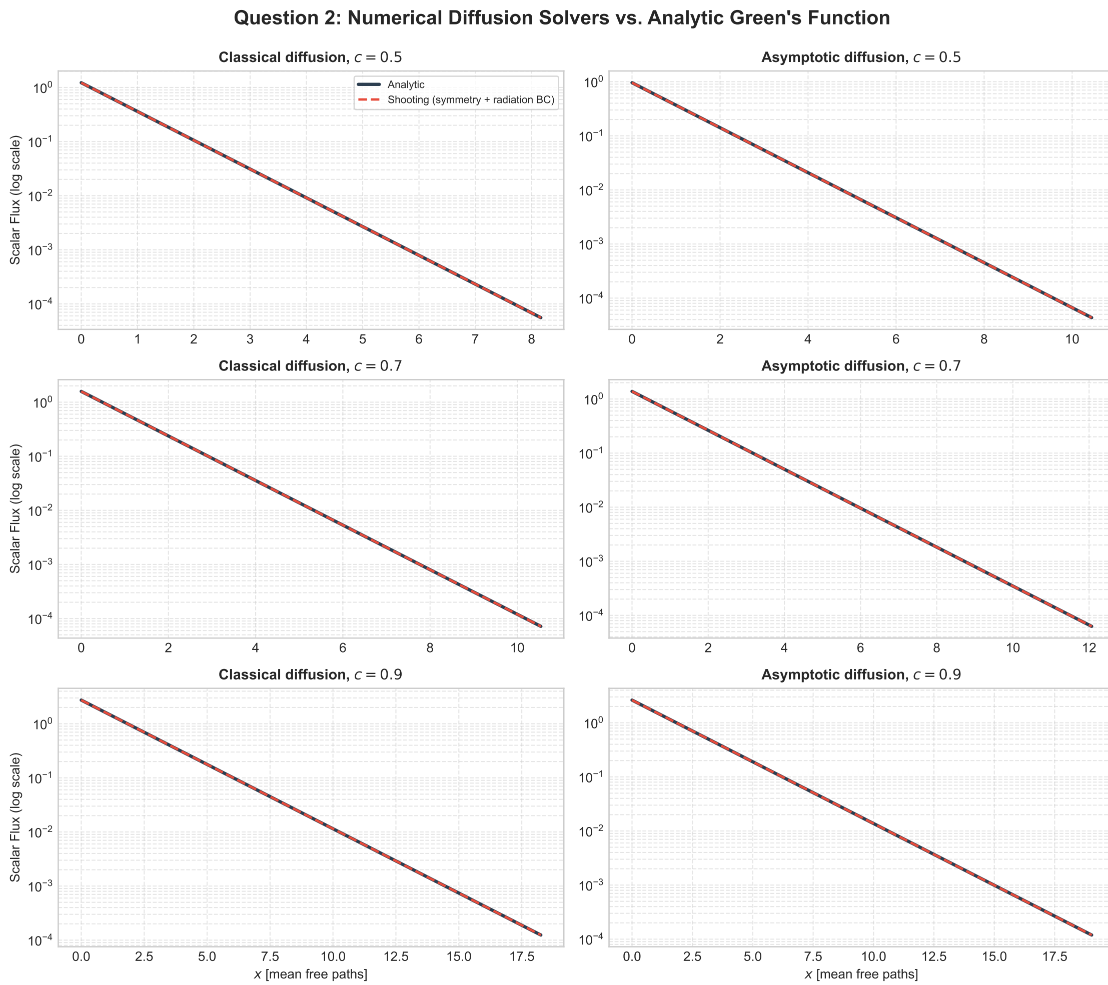

Figure 3: Numerical solution against the analytic diffusion Green's function, for both approximations at $c = 0.5, 0.7, 0.9$.

4.1 The Eigenvalue on the Multiplying Branch

For $c > 1$ the discrete eigenvalue leaves the real axis. Writing $\nu_0 = i|\nu_0|$ and $k_0 = 1/|\nu_0|$, and using $\operatorname{arctanh}(ik) = i \arctan(k)$, the transcendental equation $c\,\nu_0 \operatorname{arctanh}(1/\nu_0) = 1$ becomes real again:

$$c \arctan(k_0) = k_0 \tag{30}$$

which is the form tabulated by Case, de Hoffmann & Placzek (Table 8, Part II). The same analytic fit used in Question 1 continues onto this branch: for $c > 1$ the radicand of $\nu_0 = 1/\sqrt{1 - c^{p(c)}}$ turns negative — which is precisely the statement that ν_0 has become imaginary — so

$$|\nu_0(c)| = \frac{1}{\sqrt{c^{p(c)} - 1}}, \qquad p(c) = 2.47412 + \frac{0.00363081}{c^2} - 0.0352458\,c + \frac{0.557498}{c} \tag{31}$$

This reproduces the root of the transcendental equation to $10^{-4}\,\%$ at $c = 1.02$ and $0.10\,\%$ at $c = 2$; the transcendental root itself agrees with every entry of Table 8 Part II to 5×10^{-6}, the resolution of its printed digits.

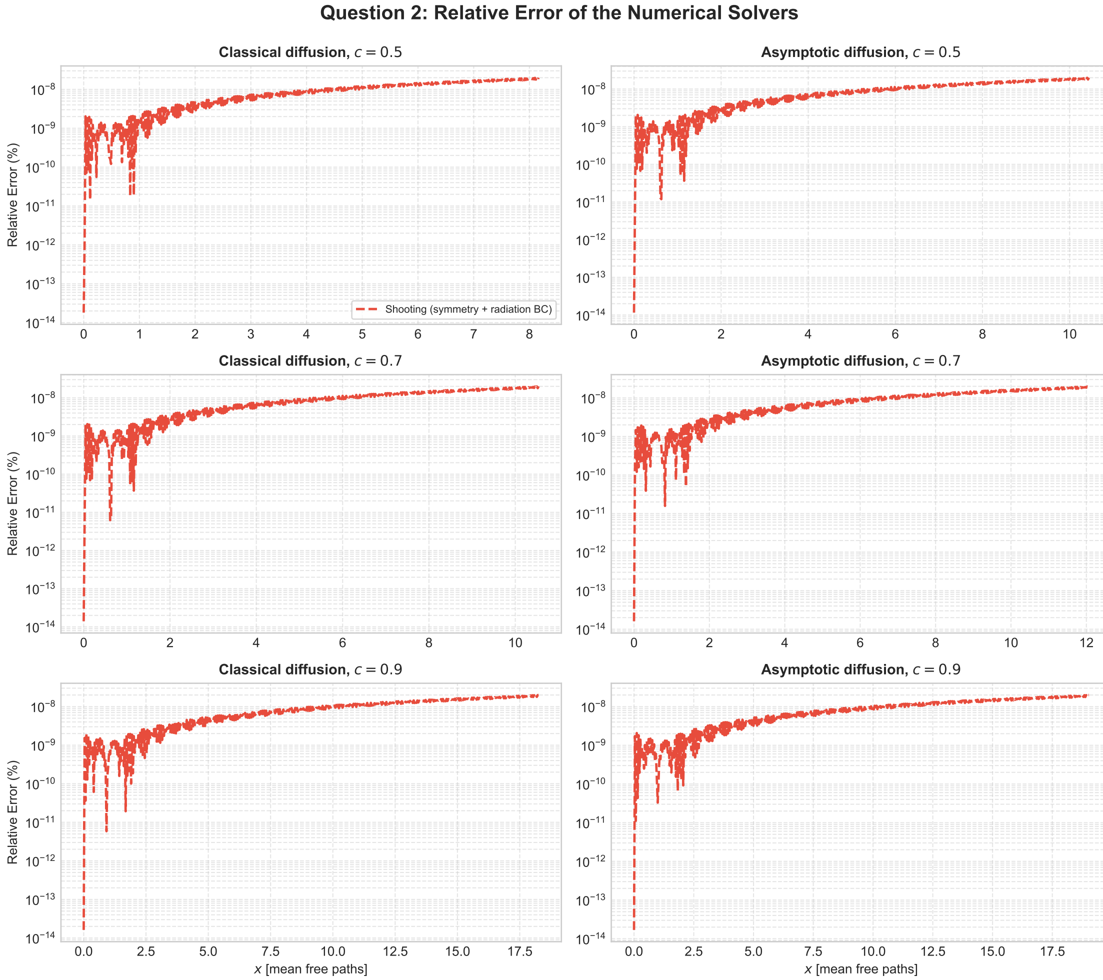

Figure 4: Relative error of the numerical solution across the domain. With the radiation boundary condition the error stays flat instead of diverging at the outer edge.

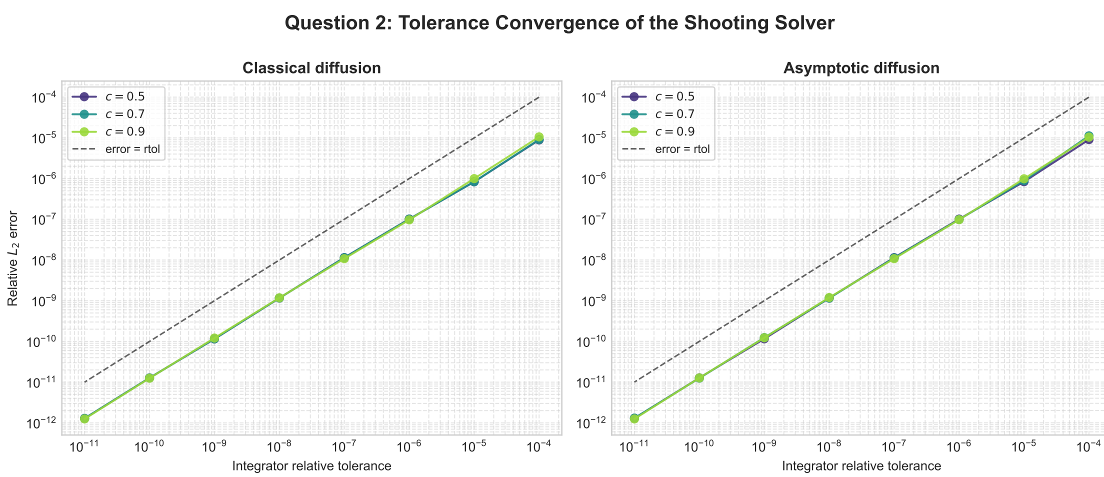

Figure 5: Tolerance convergence of the shooting solver.

An imaginary ν_0 turns the infinite-medium mode e^{-x/ν_0} into a trigonometric one, which is what makes a finite critical system possible:

$$\phi_{\text{as}}(x) = A\cos\left(\frac{\Sigma_t x}{|\nu_0|}\right), \qquad \phi_{\text{as}}(r) = A\frac{\sin(\Sigma_t r/|\nu_0|)}{\Sigma_t r} \tag{32}$$

the sine being excluded in the slab by symmetry about the mid-plane, and the cosine in the sphere by regularity at the origin.

4.2 The Criticality Relations

The asymptotic flux extrapolates to zero a distance $z_0(c)$ outside the physical surface, $\phi_{\text{as}}(-z_0) = 0$ measured from the boundary. Placing the quarter-wave zero of the cosine and the first zero (argument $= \pi$) of the sine at that point gives, in mean free paths,

$$\frac{a}{2} = \frac{\pi}{2}\,|\nu_0(c)| - z_0(c), \qquad \Sigma_t R_c = \pi\,|\nu_0(c)| - z_0(c) \tag{33}$$

These are "exact transport" in that no diffusion coefficient enters: both the eigenvalue and the extrapolation distance are transport quantities. They are still asymptotic, in that the transient part of the flux near the free surface is neglected — and for the sphere the plane Milne z_0 is used without the Winslow curvature correction, so the spherical relation is the weaker of the two for small, strongly multiplying systems.

The extrapolation distance is taken from the expansion about $c = 1$,

$$z_0(c) = 0.710446\left[\frac{1}{c} + q\,\frac{(1-c)^2}{c} + \mathcal{O}\left(\frac{(1-c)^3}{c}\right)\right] \tag{34}$$

whose leading term is exact in the sense that $0.710446/c = 0.710446\,\nu_0\,\text{arctanh}(1/\nu_0)$: all of the first-order c-dependence comes from the dispersion relation itself, and 0.710446 is Hopf's constant, the $c = 1$ Milne result.

4.3 The Sign of the Quadratic Correction

The course notes print $q = -0.0199$. Case's own Table 23, which tabulates the product $c\,z_0(c)$, shows that the correction must be *positive*: the tabulated product rises away from 0.710446 on *both* sides of $c = 1$ (Figure 7), whereas a negative q makes it fall on both sides. The magnitude is right — with $q = +0.0199$ Equation (34) reproduces the table to its four printed digits at $c = 0.9$ and $c = 1.1$ — so the printed minus sign appears to be a transcription error. Over $1.02 < c < 2$ the two signs differ by at most 2.6 % in z_0 and 3.2 % in $a/2$, with the discrepancy concentrated at the upper end: below $c = 1.2$ the choice moves $a/2$ by less than 0.07 %. Results below are quoted with the printed $q = -0.0199$; both are plotted.

4.4 Results

Both dimensions fall steeply with c: from $a/2 = 5.67$ and $\Sigma_t R_c = 12.03$ mean free paths at $c = 1.02$ to 0.33 and 1.00 at $c = 2$. A system only just above critical multiplication needs to be many mean free paths across, because $|\nu_0| \to \infty$ as $c \to 1^+$ and the flux shape flattens; at $c = 2$ the critical sphere is barely one mean free path in radius, which is also where the neglect of the transient flux is least defensible.

The error introduced by the approximate inputs is dominated by z_0, not by $|\nu_0|$: the eigenvalue fit stays below 0.11 % across the whole range, while the z_0 expansion — a two-term expansion

c	$\lvert\nu_0(c)\rvert$	$z_0(c)$	$a/2$	$\Sigma_t R_c$	error in $a/2$
1.02	4.050190	0.696510	5.665513	12.027537	0.0001 %
1.05	2.531781	0.676582	3.300331	7.277244	0.0025 %
1.10	1.756653	0.645731	2.113612	4.872955	0.0128 %
1.20	1.198272	0.591567	1.290674	3.172916	0.0667 %
1.30	0.946021	0.545518	0.940488	2.426495	0.1777 %
1.40	0.793813	0.505846	0.741072	1.987990	0.3592 %
1.50	0.689210	0.471274	0.611333	1.693941	0.6116 %
1.60	0.611746	0.440848	0.520080	1.481008	0.9535 %
1.70	0.551520	0.413834	0.452491	1.318817	1.3672 %
1.80	0.503065	0.389665	0.400547	1.190759	1.8831 %
1.90	0.463074	0.367892	0.359503	1.086897	2.4781 %
2.00	0.429411	0.348154	0.326364	1.000881	3.1859 %

Table 1: Critical dimensions in mean free paths, from the approximate $|\nu_0(c)|$ and the printed z_0 fit ($q = -0.0199$). The last column is the departure of $a/2$ from the reference evaluation, which uses the root of the transcendental equation for $|\nu_0|$ and Case's Table 23 for z_0.

about $c = 1$ evaluated as far as $c = 2$, where $(1-c)^3$ is no smaller than $(1-c)^2$ — reaches 2.6 %. Because z_0 enters both relations as an additive constant while $|\nu_0|$ is multiplied by π or $\pi/2$, the same absolute z_0 error contaminates $a/2$ about twice as strongly as R_c, and it contaminates both increasingly as the system shrinks. This is visible in the lower-right panel of Figure 6: the total error stays under 0.1 % up to $c \approx 1.25$ and grows to a few percent by $c = 2$.

4.5 Parts 3(b) and 3(c): Diffusion with Marshak and Mark Conditions

In mean free paths classical diffusion has $D = 1/3$, and a multiplying medium has a negative absorption cross section, $\Sigma_a = 1 - c < 0$. The critical equation is then a Helmholtz equation,

$$-D\nabla^2\phi + (1-c)\phi = 0 \quad\Longrightarrow\quad \nabla^2\phi + B^2\phi = 0, \qquad B^2 = \frac{c-1}{D} = 3(c-1) \tag{35}$$

whose solutions in the slab and the sphere are the *same* shapes $\cos(Bx)$ and $\sin(Br)/r$ found above. Diffusion therefore does not change the form of the criticality relations at all; it only replaces the two inputs:

$$|\nu_0(c)| \;\longrightarrow\; \frac{1}{B} = \frac{1}{\sqrt{3(c-1)}}, \qquad z_0(c) \;\longrightarrow\; \begin{cases} 2/3 & \text{Marshak} \\ 1/\sqrt{3} & \text{Mark} \end{cases} \tag{36}$$

The Marshak condition sets the incoming partial current to zero at the surface, $J^-(s) = \phi/4 + D\phi'/2 = 0$, i.e. $\phi + 2D\phi' = 0$, so $z_0 = \ell_0 = 2D = 2/3$. The Mark condition sets the incoming angular flux to zero along the P_1 quadrature direction $\mu = 1/\sqrt{3}$, giving $\phi + \sqrt{3}D\phi' = 0$ and $z_0 = \ell_0 = \sqrt{3}D = 1/\sqrt{3} \approx 0.5774$. For both, the extrapolation distance and the linear extrapolation length coincide.

The essential difference from part 3(a) is that *neither* diffusion input carries the true c-dependence: $1/B$ is only the leading term of $|\nu_0(c)|$ as $c \to 1^+$, and the two extrapolation distances are constants, whereas the transport $z_0(c) \approx 0.7104/c$ falls by a factor of two across the range.

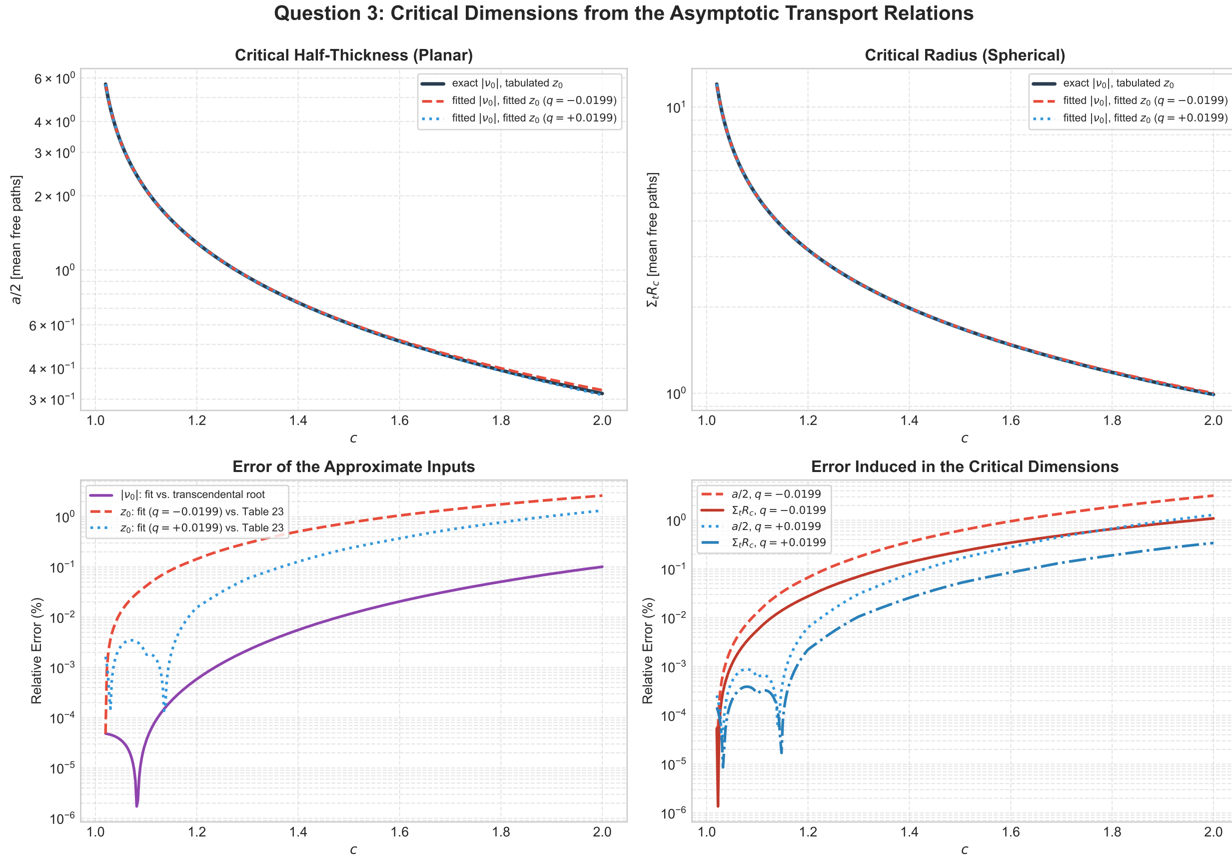

Figure 6: Critical half-thickness and critical radius for $1.02 < c < 2$ (top), and the errors of the approximate inputs and of the resulting dimensions (bottom), against a reference that uses the transcendental root for $|\nu_0|$ and Case's Table 23 for z_0.

4.6 Comparison

All three treatments agree as $c \to 1^+$ and separate as the system shrinks (Figure 8). This is the expected behaviour: near critical multiplication the system is many mean free paths across, the flux is smooth, and the boundary layer is a small correction to a large dimension; at $c = 2$ the critical sphere is barely one mean free path in radius and is boundary layer almost throughout.

Both diffusion inputs err in the same direction, but they enter the two relations with different weights, which produces qualitatively different behaviour in the slab and the sphere:

- $1/B$ *over*-estimates the relaxation length — by 0.8 % at $c = 1.02$ and 34 % at $c = 2$ — which inflates both dimensions.
- The constant z_0 *over*-estimates the extrapolation distance once c grows, since the transport $z_0(c)$ falls as $\approx 0.7104/c$ while $2/3$ and $1/\sqrt{3}$ do not. Being subtracted, this deflates both dimensions.

In the slab the relaxation length is multiplied by $\pi/2$ and in the sphere by π, so the first error is twice as strong in the sphere while the second is identical in both. The result is that Marshak's $a/2$ is non-monotonic in its error — $+5.5$ % at $c \approx 1.23$, crossing zero near $c = 1.52$, and -26 % at $c = 2$, a cancellation rather than an accuracy — whereas the spherical radius is over-estimated throughout,

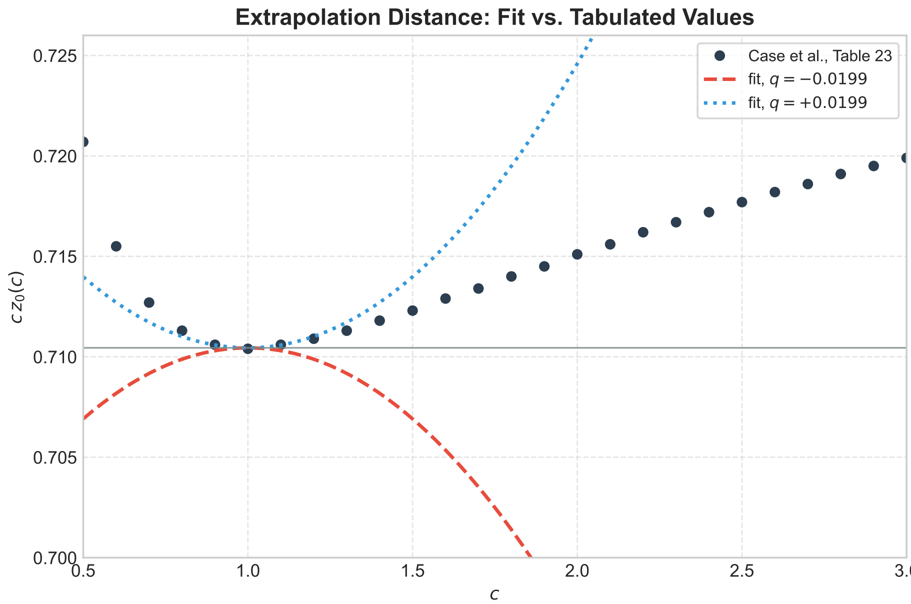

Figure 7: The tabulated product $c\,z_0(c)$ against the two-term fit with either sign of the quadratic coefficient. The tabulated values rise on both sides of $c = 1$, which fixes the sign as positive.

	$a/2$			$\Sigma_t R_c$		
c	transport	Marshak	Mark	transport	Marshak	Mark
1.02	5.665513	5.746082	5.835399	12.027537	12.158832	12.248148
1.05	3.300331	3.389112	3.478428	7.277244	7.444891	7.534207
1.10	2.113612	2.201202	2.290518	4.872955	5.069071	5.158387
1.20	1.290674	1.361223	1.450539	3.172916	3.389112	3.478428
1.30	0.940488	0.989098	1.078414	2.426495	2.644863	2.734179
1.40	0.741072	0.767268	0.856584	1.987990	2.201202	2.290518
1.50	0.611333	0.615883	0.705200	1.693941	1.898433	1.987749
1.60	0.520080	0.504136	0.593452	1.481008	1.674938	1.764255
1.70	0.452491	0.417286	0.506602	1.318817	1.501238	1.590555
1.80	0.400547	0.347278	0.436594	1.190759	1.361223	1.450539
1.90	0.359503	0.289290	0.378606	1.086897	1.245246	1.334562
2.00	0.326364	0.240233	0.329549	1.000881	1.147133	1.236449

Table 2: Critical dimensions in mean free paths from the three treatments: exact transport (3a), diffusion with Marshak (3b) and diffusion with Mark (3c) boundary conditions.

by +15 % (Marshak) and +24 % (Mark) at $c = 2$. Mark's smaller extrapolation distance makes it the larger of the two estimates everywhere, and for the slab that accidentally offsets the $1/B$ error rather well, ending at +1 % at $c = 2$; for the sphere it merely adds to it.

A final caveat on the shortcut $z_0 = \ell_0$. Setting the flux to zero at the extrapolated surface is not the same as imposing $\phi + \ell_0\phi' = 0$ on the flux shape. The latter gives, for the slab and the sphere,

$$B\,\frac{a}{2} = \arctan\!\left(\frac{1}{B\ell_0}\right), \qquad u\cot u = 1 - \frac{u}{B\ell_0}, \quad u = BR \in (0, \pi) \tag{37}$$

against $Ba/2 = \pi/2 - B\ell_0$ and $u = \pi - B\ell_0$ for the extrapolated form. Since $\arctan(1/y) = \pi/2 - y + \mathcal{O}(y^3)$, the two agree to first order in $B\ell_0$ and hence coincide as $c \to 1$: at $c = 1.02$ they differ by 0.1 % in the slab and 0.3 % in the sphere. They part company as the system shrinks — at $c = 2$ the Marshak slab gives $a/2 = 0.412$ with the condition applied and 0.240 with the extrapolated zero, a factor of 1.7 — and the applied-condition results (dotted in Figure 8) then lie above the transport answer in every case, so the good agreement of the extrapolated Marshak slab at large c really is a cancellation of two separate errors.

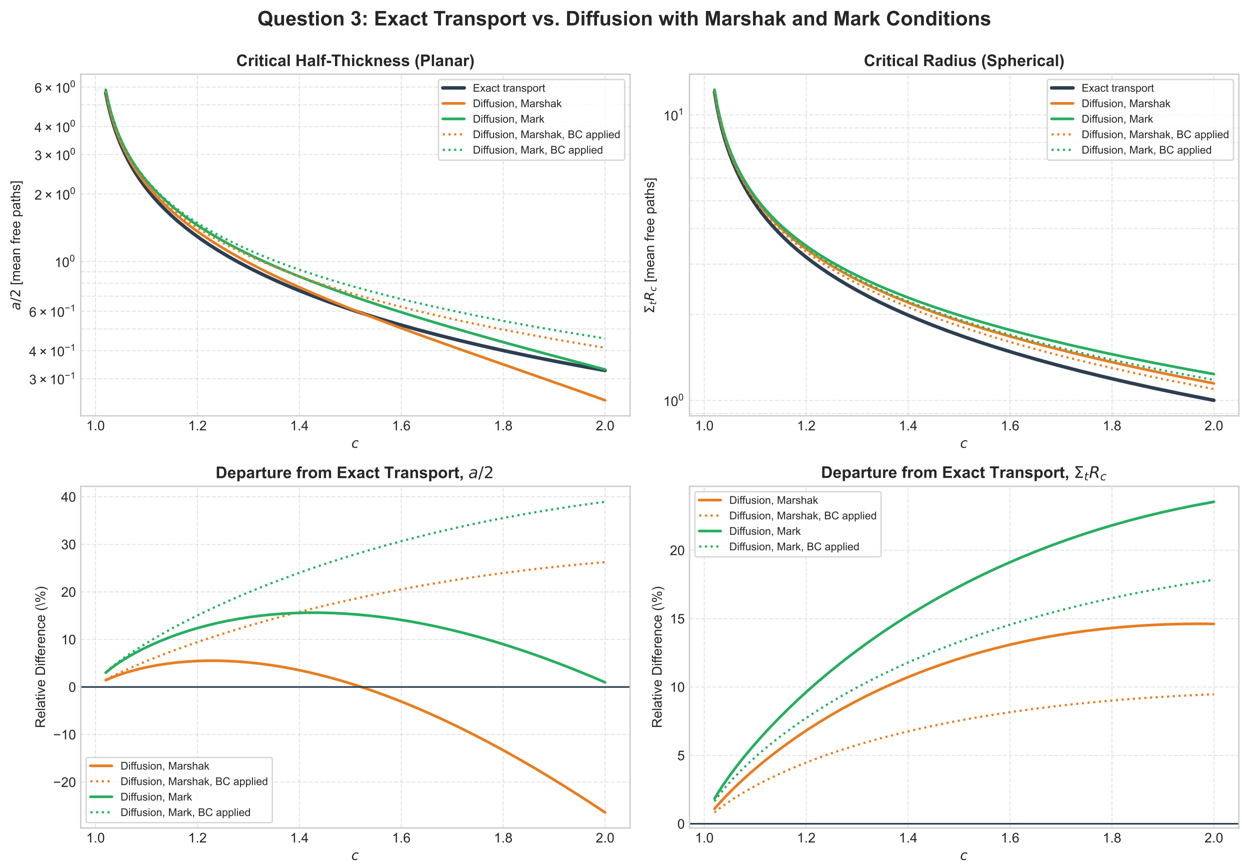

Figure 8: Exact transport against diffusion with Marshak and Mark boundary conditions (top), and the signed departure of each diffusion result from the transport one (bottom). Dotted curves impose the boundary condition on the flux shape instead of using an extrapolated zero.

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5 Question 4: Numerical Diffusion Code in Spherical Coordinates

The code solves the steady-state diffusion equation with a fission source in a bare sphere,

$$-D\frac{1}{r^2}\frac{d}{dr}\left(r^2\frac{d\phi}{dr}\right) + \Sigma_a\phi(r) = \frac{1}{k}\nu\Sigma_f\phi(r) \tag{38}$$

and finds the radius at which $k = 1$. Dividing the fission source by k makes a solution exist at every radius, so criticality becomes a root-finding problem on R rather than an eigenvalue search. Both approximations use the same solver and differ only in two constants:

$$\text{classical:}\quad D = \frac{1}{3\Sigma_t},\quad z_0 = 2D,\qquad \text{asymptotic:}\quad D = \frac{(c-1)|\nu_0(c)|^2}{\Sigma_t},\quad z_0 = z_0(c) \tag{39}$$

The asymptotic coefficient is the continuation above $c = 1$ of the $D = (1-c)\nu_0^2$ of Question 2: with $\nu_0 = i|\nu_0|$ both ν_0^2 and $1-c$ change sign, so D remains positive. Its purpose is to give the diffusion equation the true transport relaxation length, $1/B = |\nu_0|/\Sigma_t$, in place of the classical $1/\sqrt{3(c-1)}$.

5.1 Discretisation

The equation is discretised by finite volume on N uniform cell-centred shells. Integrating (38) over cell i turns the divergence term into a difference of surface currents,

$$A_{i+1/2}J_{i+1/2} - A_{i-1/2}J_{i-1/2} + \Sigma_a V_i\phi_i = S_i V_i,\qquad J_{i+1/2} = -D\,\frac{\phi_{i+1}-\phi_i}{h} \tag{40}$$

with $A = 4\pi r^2$ and $V_i = \frac{4\pi}{3}(r^3_{i+1/2} - r^3_{i-1/2})$. Writing the balance rather than the derivatives is what removes the $1/r^2$ singularity at the origin: the innermost face has $A_{1/2} = 0$, so the term vanishes on its own and the symmetry condition $\phi'(0) = 0$ is imposed exactly, with no special case in the code. The resulting matrix is tridiagonal and is solved in $O(N)$.

Both outer boundary treatments are the single condition $\phi + \ell_0\phi' = 0$. Eliminating the surface flux gives one leakage coefficient,

$$J_s = \frac{D\,\phi_{N-1}}{\ell_0 + h/2} \tag{41}$$

which covers the extrapolated zero ($\ell_0 = 0$ on a mesh run out to $R + z_0$) and the Robin condition ($\ell_0 = z_0$ at the physical surface R) without a branch. The extrapolated form is used below, since it is what the analytic relations of Question 3 assume, making the comparison a test of the numerics alone.

5.2 The Iterative Algorithm

k is obtained by the source iteration of Bell & Glasstone, pp. 189–192:

1. start from a flat flux $\phi^{(0)}$ and $k^{(0)} = 1$;
2. form the fission source $S^{(n)} = \frac{1}{k^{(n)}}\nu\Sigma_f\phi^{(n)}$;
3. solve the removal problem $\left(-D\nabla^2 + \Sigma_a\right)\phi^{(n+1)} = S^{(n)}$, one banded solve;
4. update k by the ratio of successive fission-source integrals,

$$k^{(n+1)} = k^{(n)}\,\frac{\int \nu\Sigma_f\phi^{(n+1)}\,dV}{\int \nu\Sigma_f\phi^{(n)}\,dV} \tag{42}$$

5. repeat until $|k^{(n+1)} - k^{(n)}| < 10^{-10}k^{(n+1)}$.

The flux is renormalised to unit maximum after each sweep. Each sweep multiplies its amplitude by roughly k, and over the several hundred sweeps this problem needs the iterate would otherwise drift by many orders of magnitude and the convergence test would lose its significant digits. The rescaling is physically irrelevant because the ratio in step 4 is homogeneous in ϕ.

Convergence is at the dominance ratio $k_1/k_0 = (\Sigma_a + DB_0^2)/(\Sigma_a + DB_1^2)$, which tends to unity for a large weakly multiplying sphere — about 0.94 at $c = 1.02$, so several hundred sweeps are needed there. The critical radius is then found by bisection (Brent) on $k(R) - 1$, which is monotonically increasing in R, bracketed within $\pm 5\%$ of the analytic radius so that k is never evaluated far from criticality.

5.3 Comparison with the Analytical Result

At $k = 1$ equation (38) reduces to a Helmholtz equation with buckling $B^2 = (\nu\Sigma_f - \Sigma_a)/D = \Sigma_t(c-1)/D$, whose regular solution $\sin(Br)/r$ vanishes at the extrapolated radius when

$$R_c = \frac{\pi}{B} - z_0 \tag{43}$$

Substituting the two parameter sets recovers exactly the relations of Question 3: $\Sigma_t R_c = \pi/\sqrt{3(c-1)} - 2/3$ for classical diffusion with Marshak conditions, and $\Sigma_t R_c = \pi|\nu_0| - z_0(c)$ for asymptotic diffusion, the latter being the exact-transport relation of part 3(a). Note that R_c depends on the cross sections only through c, since Σ_a and $\nu\Sigma_f$ enter (43) only as their difference.

Table 3 compares the two. On a 400-cell mesh the numerical radius agrees with (43) to between $1.6 \times 10^{-4}\,\%$ and $2.4 \times 10^{-4}\,\%$ over the whole range $1.02 < c < 2$, for both approximations.

Table 3: Critical radius in mean free paths from the $k = 1$ search ($N = 400$) against the analytic result (43).

	Classical		Asymptotic		
c	numerical	analytic	numerical	analytic	difference (classical)
1.02	12.15881	12.15883	12.02752	12.02754	$-1.6 \times 10^{-4}\,\%$
1.10	5.06906	5.06907	4.87295	4.87295	$-1.7 \times 10^{-4}\,\%$
1.20	3.38911	3.38911	3.17289	3.17289	$-1.8 \times 10^{-4}\,\%$
1.50	1.89843	1.89843	1.69369	1.69369	$-2.1 \times 10^{-4}\,\%$
1.80	1.36122	1.36122	1.18995	1.18996	$-2.3 \times 10^{-4}\,\%$
2.00	1.14713	1.14713	0.99952	0.99952	$-2.4 \times 10^{-4}\,\%$

The residual is discretisation error alone. The scheme is second order: refining the mesh at $c = 1.5$ drives the relative error of R_c from 5.28×10^{-4} at $N = 25$ to 5.16×10^{-7} at $N = 800$, a factor of exactly 4.00 per doubling at every step (Figure 10). The sign is consistently negative — the discrete operator is slightly too leaky on a coarse mesh and so reaches $k = 1$ at a marginally smaller radius. Two further checks: k at the returned radius equals 1 to ten digits, and the converged flux matches the fundamental mode $\sin(Br)/Br$ to 3.2×10^{-6} across the whole mesh (Figure 11).

The two approximations differ from each other far more than either differs from its own analytic solution, and the gap is entirely in the relaxation length. Classical diffusion gives the larger radius throughout, by $1.1\,\%$ at $c = 1.02$ rising to $14.8\,\%$ at $c = 2$, because $1/\sqrt{3(c-1)}$ is only the $c \to 1$ limit of $|\nu_0(c)|$.

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6 Question 5: Bare Critical Mass of U-235 and Pu-239

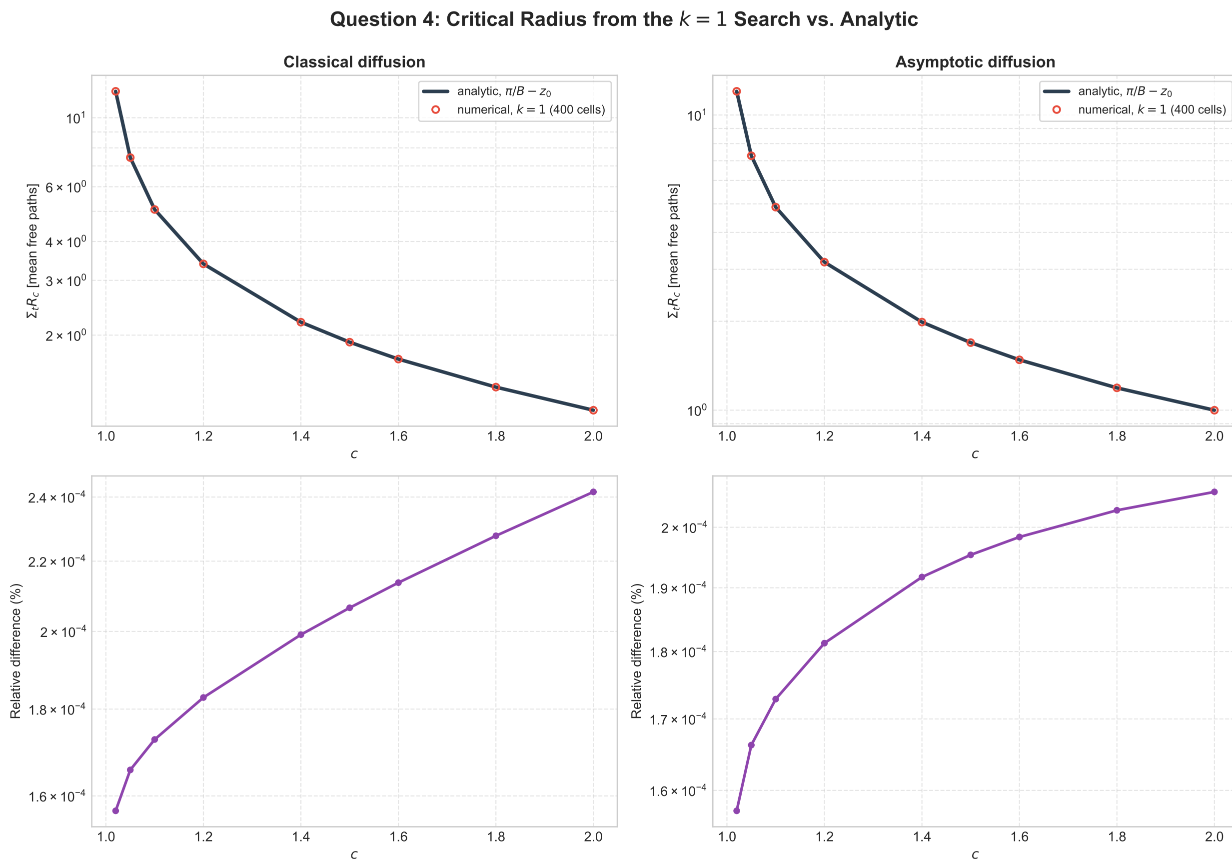

Figure 9: Critical radius from the $k = 1$ search (markers) against the analytic radius $\pi/B - z_0$ (solid), for both diffusion approximations, with the relative difference below.

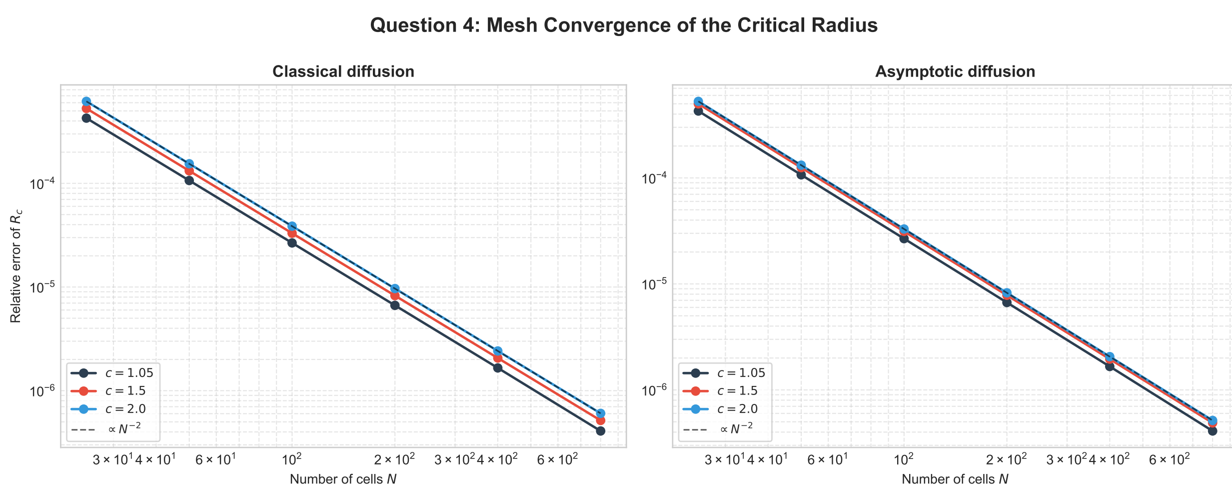

Figure 10: Mesh convergence of the critical radius. The error follows the second-order guide $\propto N^{-2}$ at every refinement.

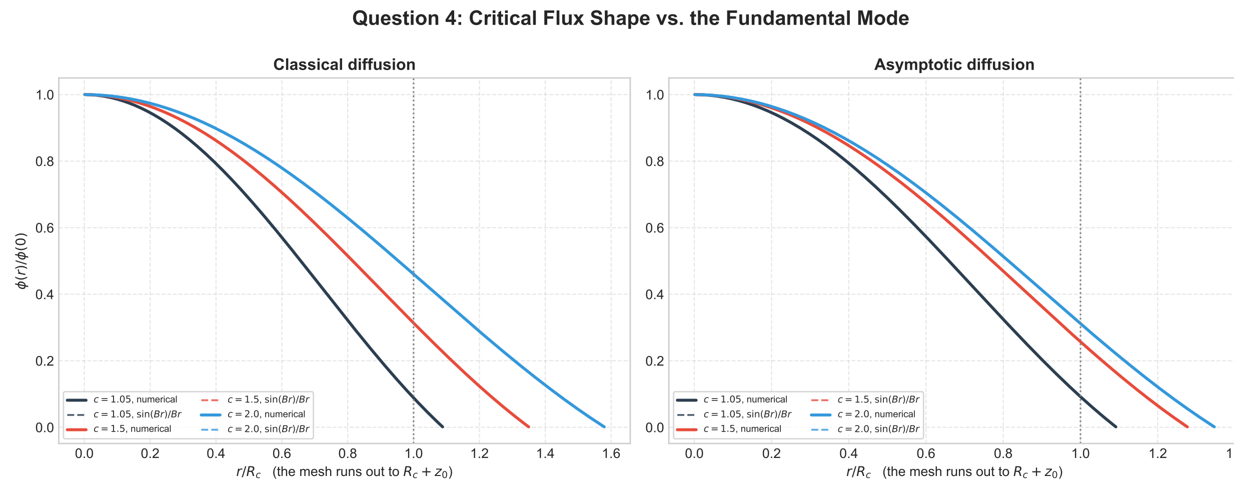

Figure 11: Converged critical flux (solid) against the fundamental mode $\sin(Br)/Br$ (dashed), normalised at the centre. The mesh continues past $r/R_c = 1$ to the extrapolated radius $R_c + z_0$, where the flux is driven to zero.

Question 5 is Question 4 run with the one-group benchmark cross sections of Sood, Forster & Parsons (2003), with the mass following from the radius as $M_c = \rho \frac{4}{3}\pi R_c^3$. Both fissile materials have $\Sigma_t = 0.32640\ \text{cm}^{-1}$, a mean free path of 3.0637 cm, so both critical spheres are only about two mean free paths in radius.

Table 4: Bare critical radius and mass, from the $k = 1$ search on a 400-cell mesh. Numerical and analytic radii agree to four decimal places in cm.

Material	Approximation	c	ρ [g/cm^3]	R_c [cm]	$\Sigma_t R_c$	M_c [kg]
Pu-239	classical	1.50	15.7	5.8163	1.8984	12.940
Pu-239	asymptotic	1.50	15.7	5.1890	1.6937	9.188
U-235	classical	1.30	19.0	8.1031	2.6449	42.345
U-235	asymptotic	1.30	19.0	7.4339	2.4264	32.696

Because the mass goes as the cube of the radius, the moderate difference in relaxation length between the two approximations is strongly amplified. The classical radius exceeds the asymptotic one by 12.1 % for Pu-239 and 9.0 % for U-235, which becomes a mass ratio of 1.41 and 1.30 respectively — classical diffusion over-predicts the critical mass by 30–40 % on these materials, and the discrepancy grows with c. The asymptotic values are the ones to quote, since asymptotic diffusion with the exact $z_0(c)$ reproduces the exact-transport criticality relation.

These are one-group, isotropic-scattering results and should not be read as predictions of real critical masses. With a critical radius of only about two mean free paths, the boundary layer occupies a large fraction of the system and diffusion theory of either flavour is being used well outside its regime; the published transport benchmark radii of Sood *et al.* are the appropriate reference against which to judge them.

A note on the U-235 data. Two U-235 rows were in circulation for this work. The assignment table gives $\Sigma_c = 0.013056$, $\Sigma_s = 0.248064$, $\Sigma_t = 0.32640$, whose own cross sections yield $c =$

$(\nu\Sigma_f + \Sigma_s)/\Sigma_t = 1.3000$, matching the quoted c exactly; these are the values used above. An alternative row ($\Sigma_c = 0.015672$, $\Sigma_s = 0.180448$, $\Sigma_t = 0.26140$) is quoted with $c = 1.50$ but in fact implies $c = 1.3646$, and so is not self-consistent. Run through the same code it gives $R_c = 8.9412$ cm and $M_c = 56.89$ kg classically, or $R_c = 8.1161$ cm and $M_c = 42.55$ kg asymptotically — roughly 30 % more mass. The Pu-239 row is identical in both sources and is unaffected.

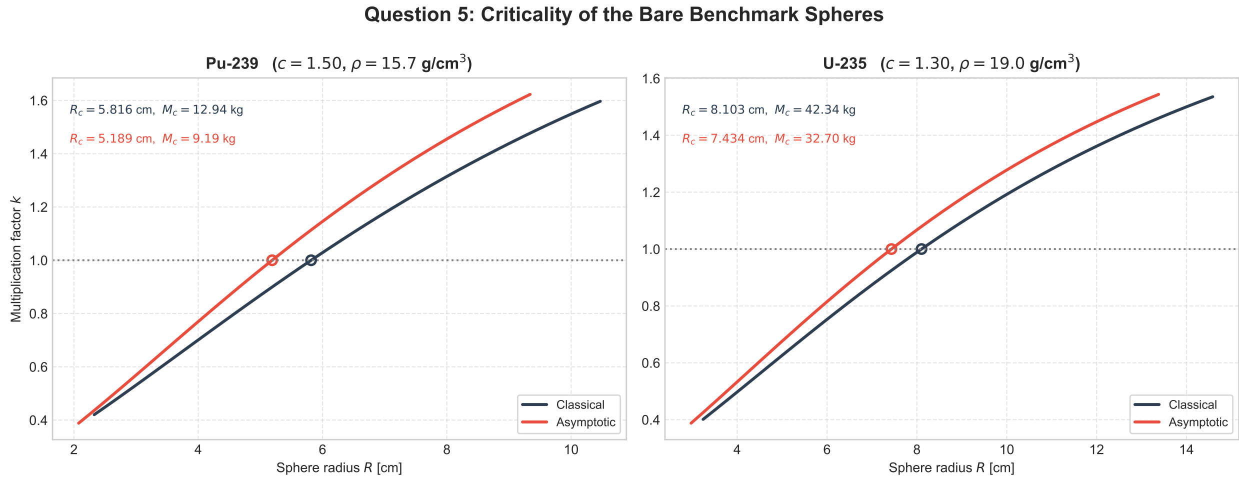

Figure 12: Multiplication factor against sphere radius for the two fissile benchmark materials, under both diffusion approximations. The circles mark the $k = 1$ crossing that defines the critical radius.